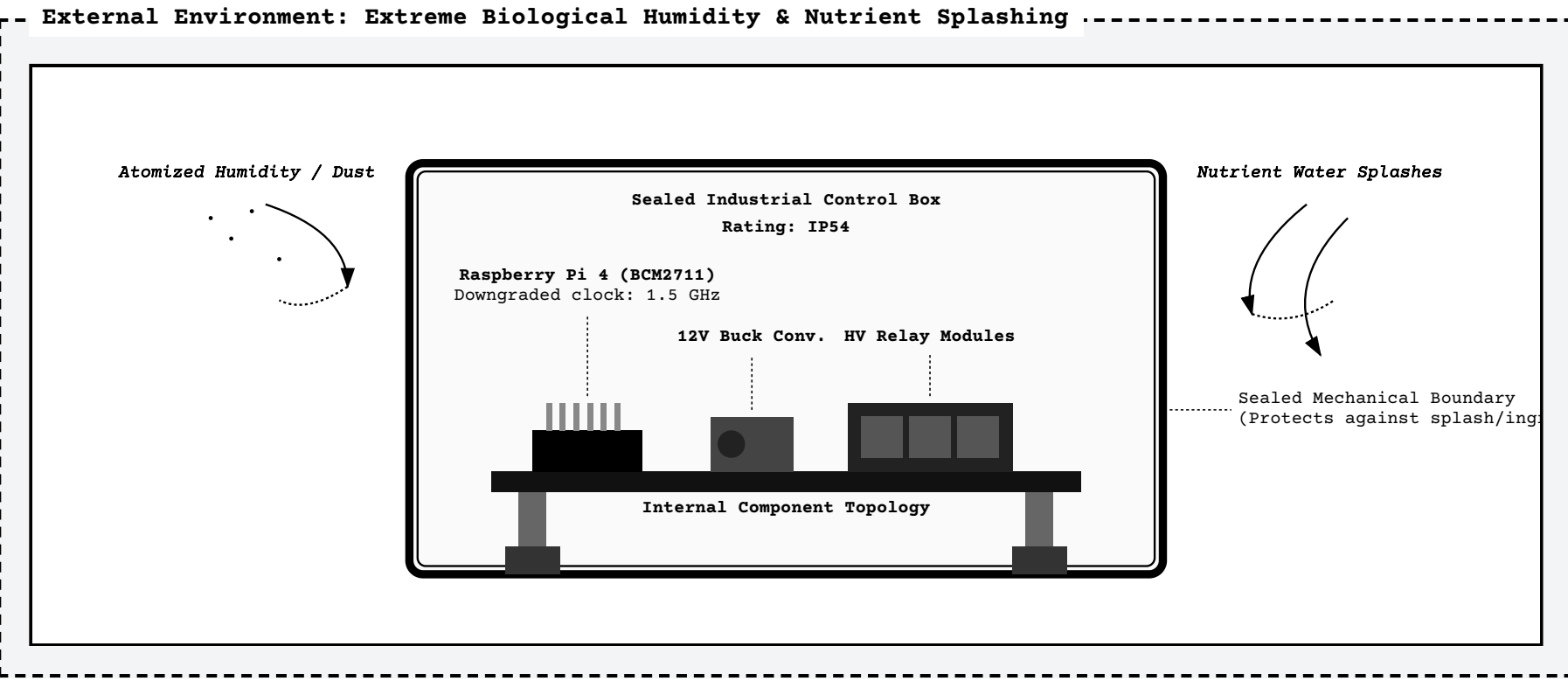


FIGURE 2: ARCHITECTURAL PIVOT
IP54 Environmental Isolation & BCM2711 (Pi 4) Thermal Mitigation Strategy



STRATEGIC COUNTERMEASURES EXECUTED

STRATEGY I	STRATEGY II	STRATEGY III		
Thermal Downgrade SoC Replacement: Transitioned from Pi 5 to Raspberry Pi 4 Model B (BCM2711). Clock Speed Reduction: Base clock reduced from 2.4 GHz to 1.5 GHz. Net Effect: Substantially lower localized thermal output, eliminating the sharp temperature differential causing micro-condensation.	IP54 Sealing Topology Change: Exposed breadboard topology entirely abandoned. Ingress Protection (IP): Encased in industrial box rated IEC 60529 IP54. <table><tr><td>5 = Robust Dust Protection</td><td>4 = Multi-directional Splash Guard</td></tr></table>	5 = Robust Dust Protection	4 = Multi-directional Splash Guard	Physical Isolation Custom Engineering: 3D-modeled standoffs designed to elevate computational logic. Mechanical Shielding: Physically isolates sensitive PCB contacts from accidental pooling or bottom-up condensation. RESULT: ZERO SUBSEQUENT SBC FAILURES
5 = Robust Dust Protection	4 = Multi-directional Splash Guard			